♡ cast of characters ♡

capacitors
more capacitance, more chonk
(the wires go in one end and out the other)
10pF 1nF 100nF 1µF 4µ7F 22µF

resistor
grumpy, throws bricks
(and the odd molotov)

ferrite bead
a very hungry inductor

crystal
dances at its own frequency ♪

FPGA
Stimulus & Sampling Pulse Timing Generation

U5 GW1NZ-LV1QN48C6

Pulse Amplifier

U2 PI3HDX12311

Signal Splitter
Couples incident stimulus into mesh, reflections into sampler

Sampling Bridges

BAT15-04W

Place clip line close to diodes

Low-Speed Amplifiers

U13A OPA4354
U13B
U13C
U13D

L/R Connector Mux

U10 TMUX1574

Termination Mux & Crossbar
Selects termination, and flips mesh send/return

U4 TMUX5446
U12 TMUX5446

Sub-Mesh Output Mux

U5 PI3WVR41310
U9 PI3WVR41310

TODO: ESD diodes here?

J1 Conn_01x30_MountingPin
J2 Conn_01x30_MountingPin

Microcontroller

U15 STM32U575RGTx
Y1 Crystal_GND24_Small

Debug USB

J5 USB_C_Receptacle_USB2.0_16P
U16 USBLC6-4SC6

Debug connector

J3
Wuerth WR-WST Redfit IDC

PiHAT+ Connector

HAT+ ID EEPROM
U11 M24C32-FMN
J4 Raspberry_Pi_4

Power Supplies
5V input, outputs: 3V3D (digital), 3V3A (analog, low noise), 1V8 (digital), 1V2 (digital)

U7 TH7003C
U8 TLV74333PDBVR
U6 TLV74312PDBVR
U17 TPS51200DRC

let's sense some pillows~! ♡

✦ センサー回路図 ✦

yanartas-pillowbox-sensors

~ the pillowbox sensor board ~

sheet ♡ 1 / 1

KiCad 10 · paper A1 · drawn with love ♪